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# Geomechanical Evolution of the South-West Hub Carbon Capture and Storage Reservoir in Western Australia

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## Abstract

Characterising the mechanical behaviour of porous subsurface reservoirs is essential for avoiding hazards such as wellbore failure, fault reactivation, seal leakage and gas migration during $CO_2$ storage. Reactivation of faults is a particular concern when $CO_2$ is injected into reservoirs that contain them. To address this, a geomechanical model (GM) was built for the South-West Hub (SWH) Carbon Capture and Storage (CCS) reservoir, drawing on data from the Harvey field wellbores in Western Australia. The 3D GM combined a geological model with seismic data, dynamic-simulation output, and one-dimensional mechanical-earth models. A 4D finite-element model was then used to capture how $CO_2$ injection alters stress and deformation within the storage formation, and to characterise the failure behaviour and performance of the reservoir as injection proceeds. Effective-stress azimuth and magnitude were tracked throughout the injection period. Estimated magnitudes for $S_v$, $S_H$ and $S_h$ across the study area show $S_H$ ranging from 22 to 75 MPa and $S_h$ from 21 to 72 MPa.

Fault strain responses are compared across a set of injection scenarios. Five separate reservoir-pressure and saturation profiles were used to calibrate the 3D model, which was then re-run repeatedly to refine the stress model against available leak-off test (LOT) data. The calibrated model was subsequently used to run fault-stability analyses and identify which faults are most prone to reactivation under the pore-pressure changes generated by ongoing $CO_2$ injection. Over 30 faults intersecting the reservoir were mapped, mostly trending N-S in keeping with the dominant regional fault orientation. Combining the dynamic and geomechanical results indicates that $CO_2$ injection does not pose a substantial fault-failure risk for the SWH reservoir. This assessment describes the isothermal, pressure-driven response: thermal stresses arising from injecting $CO_2$ colder than the formation are not represented and may be significant near the injection wells.

**Keywords:** 4D Finite-element analysis, coupled geomechanical modeling, fault reactivation risk, in-situ stresses, $CO_2$ sequestration.

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## Highlights

- Making the geomechanical model (GM) of the South West Hub Carbon Capture and Storage (CCS), Western Australia by integration of geological, geomechanical, and dynamic reservoir models.
- Analyzing rock deformation due to $CO_2$ injection scenarios from nine injection wells
- Determining the structures most susceptible to reactivate due to induced pore pressures after $CO_2$ injection
- There are no signs of fault activation based on the 4D geomechanical model
- Weak faults might bring about yield adjacent to the faults.

## Introduction

Global warming is a major challenge that will require sustained effort over coming decades to address. In response, researchers have pursued ways to cut $CO_2$ emissions from fossil-fuel use and to limit the longer-term consequences of a warming climate (Bachu 2000; Preston and Jones 2006; Bhattacharyya et al., 2021). Among the key technologies for doing so is capturing surplus $CO_2$ and injecting it into underground porous rock formations, such as depleted petroleum reservoirs (Herzog et al., 1998; Bachu and Adams., 2003; Matter and Takahashi., 2007; Steefel et al., 2013; Jia et al., 2019; Gin et al., 2021).

Certain Australian sub-surface reservoirs, among them the Harvey Ridge in south-western Australia, appear well suited to large-volume $CO_2$ storage (Olierook et al., 2013; Urosevic et al., 2018). A comparatively low natural pore pressure, resulting from limited formation fluid, is one of the key indicators used to screen candidate $CO_2$-storage reservoirs (Harpalani et al., 2006; Abdulkarem et al., 2012; Liu et al., 2015). As injection proceeds, however, rising pore pressure shifts the effective stress state of the formation, raising the chance of fault reactivation and/or seal leakage near the reservoir. $CO_2$ volumes therefore need to be kept below the level that would push pore pressure past the threshold at which faults reactivate or the seal leaks. Defining that critical pore-pressure threshold gives the storage reservoir a margin of safety that helps manage these risks (Mazumder et al., 2006; Anggara et al., 2016; Sampath et al., 2020). Coal seams represent another potential storage target with distinct injectivity and containment behaviour, as reviewed by Hosseinzadeh et al. (2024a).

Site characterisation is the first stage of any $CO_2$-injection project (Doughty et al., 2008; Dai et al., 2014; Harp et al., 2017; Yang et al., 2010; Birkholzer et al., 2014; Chen et al., 2020). Geomechanical modelling offers an early, practical way to gauge whether a candidate reservoir is suitable, revealing the mechanical characteristics of the reservoir and its cap rock and flagging any susceptibility to fracturing. Quantified model outputs support the long-term safety case for injection by indicating where and what type of pore-pressure monitoring is needed, and by giving a systematic basis for accountability over the life of the project (Al-Ameri et al., 2016; Lyu et al., 2018; Liu et al., 2021). This information, together with ongoing formation monitoring, can also help improve the technical and economic performance of the reservoir over time. A recent review by Khan et al. (2024) brings together the numerical approaches used across the field for modelling reservoir pressure, ground uplift, caprock fault reactivation, and CO2 leakage, together with the monitoring techniques used to verify stored CO2. Complementary approaches that infer reservoir pressure directly from surface deformation data offer a lower-cost alternative to dense subsurface instrumentation for this kind of monitoring (Abdollahi et al., 2024).

A clear picture of reservoir–seal efficiency, and of how durable that seal remains as formation pressure climbs, is needed to demonstrate transparently and accountably that $CO_2$ injection can continue safely without leakage (Frash et al., 2017; Palusznet al., 2020). An integrated geomechanical modelling workflow, drawing together detailed reservoir geology, the in-situ stress regime, rock mechanical properties and pore-pressure data, gives a fuller picture of how the reservoir will behave as stress conditions evolve. Such

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an approach can be used to assess and manage geomechanics-related hazards, including instability of the borehole and reservoir during drilling and injection, reservoir stability throughout the injection period, fault sealing and (re)activation linked to stress changes, and the integrity of the seal rock (Gor et al., 2013; Espinoza et al., 2017; Zappone et al., 2021). One route to these assessments is the mechanical earth model (MEM), a mathematical description of a reservoir's elastic rock parameters, stress measurements and pore pressures (Amiri et al., 2019; Noohnejad et al., 2021; Sun et al., 2021). A one-dimensional (1D) MEM feeds into the construction of a three-dimensional (3D) reservoir geomechanical model, from which numerical simulations of caprock integrity can be generated (Olabode et al., 2013; Karimnezhad et al., 2014; Yin and Ding., 2019; Raziperchikolaee et al., 2020). A similarly configured 4D coupled geomechanical approach has been used elsewhere to evaluate fault reactivation and surface deformation during reservoir depletion and CO2 sequestration (Hosseinzadeh et al., 2024b), and to assess formation failure and fault reactivation during hydrogen storage and production (Hosseinzadeh et al., 2025b). More recently, machine-learning surrogates trained on Mohr-Coulomb-based geomechanical criteria have been proposed as a faster alternative to full numerical simulation for forecasting fault-reactivation risk during CO2 storage (de Lima et al., 2026).

This study develops a 3D coupled reservoir geomechanical model to evaluate reservoir–seal integrity for the proposed storage of $CO_2$ in the Harvey Ridge reservoir, Western Australia. The model draws on a wide range of compiled data, including reservoir geology, geomechanical parameters, mapped faults and fractures, petrophysical well logs, interpreted water saturation, and reservoir pressures taken from a dynamic simulation model, with seismic inversion providing a further route to constraining reservoir elastic and petrophysical properties (Hosseinzadeh et al., 2025a). Initial pore-pressure distribution is calculated using the corrected Eaton (1975) equation, while in-situ stress values are derived through a poroelastic approach and calibrated against leak-off test (LOT) and extended leak-off test (XLOT) data. Borehole image (BHI) log interpretation is used to establish the prevailing horizontal in-situ stress directions in the reservoir, and because the reservoir's stress state shifts as $CO_2$ is injected, a pore-pressure–stress coupling approach is applied across the different injection phases considered (Vidal et al., 2009; Kim and Hosseini., 2014; Khan et al., 2018; Konstantinovskaya et al., 2020; Jung et al., 2020). Once built and calibrated, the 3D MEM is coupled with pressure and water-saturation data to yield outputs that indicate how well the existing fault system will hold and how susceptible it is to reactivation as pore pressure rises.

## Geological setting

### Tectonics

The Perth Basin is a north-south-trending regional extensional feature that lies along the margin of Western Australia's coastline. It formed through Permian–Cretaceous tectonic activity linked to the separation of Australia from India (Harris., 1994; Mory et al., 2005; Dillinger and Esteban., 2014). A Permian–Triassic extension event drove the complex rifting seen in the area and produced substantial clastic sediment deposits during that time. Extension continued into the Jurassic, accompanied by the development of graben structures that were infilled mainly with marine sediments. This rift system later contributed to the break-up of the Gondwana supercontinent, generating dextral trans-tensional stresses across the basin's northwest during the Late Cretaceous and producing widespread tectonic deformation ranging from reverse and strike-slip faulting to volcanism and erosion (Plumb, 1979; Song and Cawood, 2000; Borissova et al., 2014; Hall et al., 2017; Olierook et al., 2019).

To the east, the basin is bounded by the Yilgarn Craton and the north-south-trending Darling Fault, while to the west it extends offshore towards the coastal margin. Listric faults trending north-northwest dominate the basin and governed sediment deposition, which broadly kept pace with the accommodation space created by rift-related faulting and subsidence (Lambeck., 1987; Whittaker et al., 2013; Olierook et al., 2015; Cai et al., 2017).

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The study area sits on the Harvey Ridge, within the onshore southern Perth Basin. This ridge developed mainly from northwest–southeast-trending transfer activity (Langhi et al., 2014) (Fig. 1). The portion of the Harvey Ridge covered by the static and dynamic models includes four Harvey Field wellbores, shown in Fig. 2.

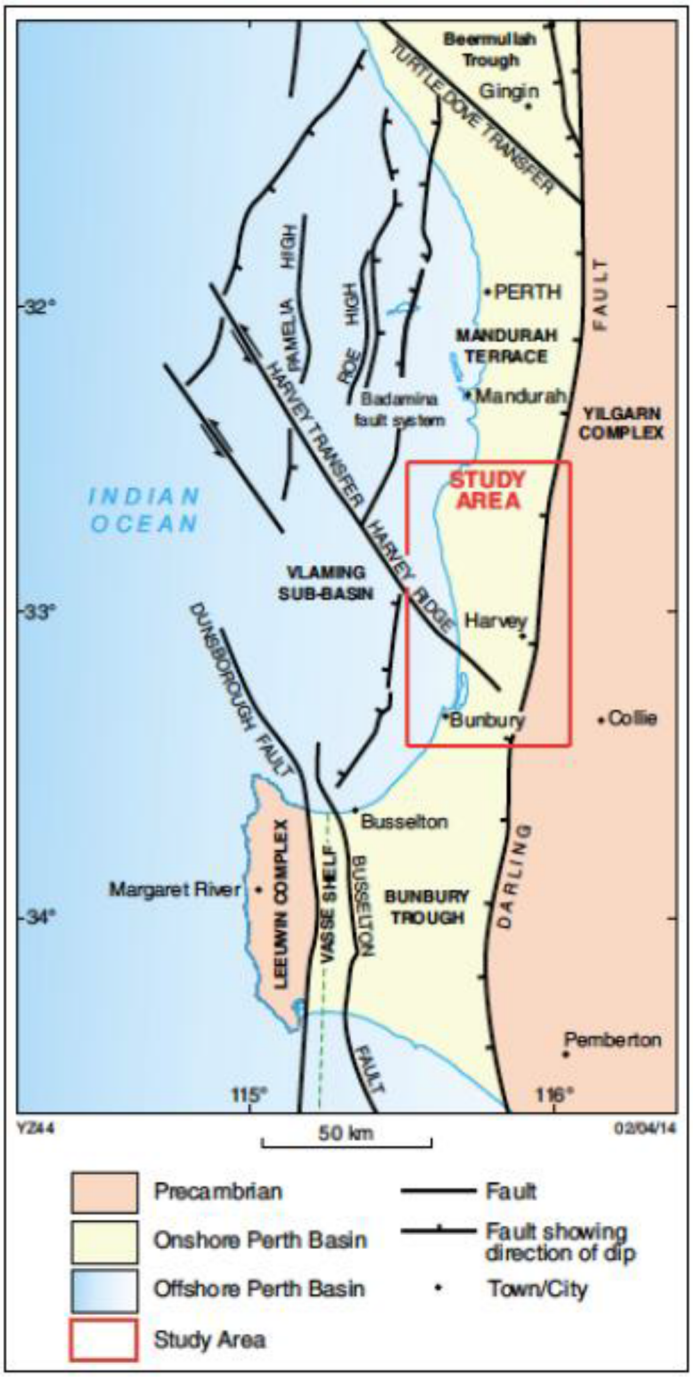



**Figure 1—Regional location map of the Perth Basin identifying the studied area. Modified after Langhi et al. (2014).**

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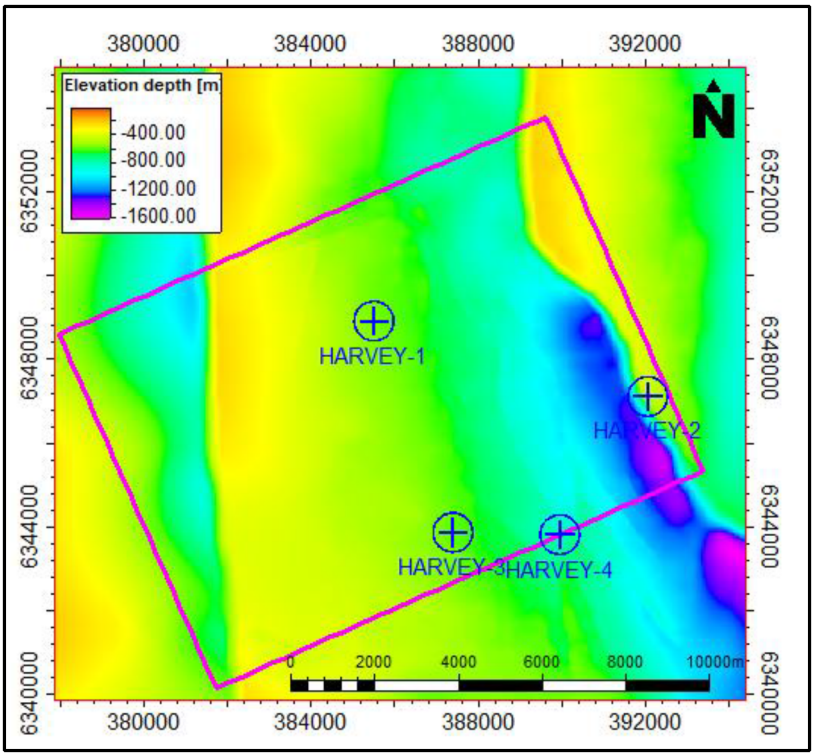



**Figure 2—Location of GeoGrid area identifying the location of the four existing Harvey wellbores. The background image is the surface map of Basal Eneabba Formation.**

## Stratigraphy

The Perth Basin's sedimentary sequence spans from the Permian through to the Quaternary (Fig. 3). Within the Triassic–Jurassic succession sit the Sabina Formation (averaging 165 m thick) and the Lesueur Sandstone (about 2070 m thick). The Lesueur Sandstone is split into a lower 'Wonnerup' member and an upper 'Yalgorup' member (Crostella and Backhouse., 2000; Mory et al., 2005; Ferdinando and Longley., 2015; Platford., 2016): the Wonnerup is largely homogeneous sandstone, while the Yalgorup contains sandstone interbedded with shale. Overlying the Lesueur Sandstone is the Eneabba Formation, whose basal shale forms the cap rock for the Lesueur sandstone reservoir, as identified in the Harvey-3 and Harvey-4 wellbores (Fig. 4).

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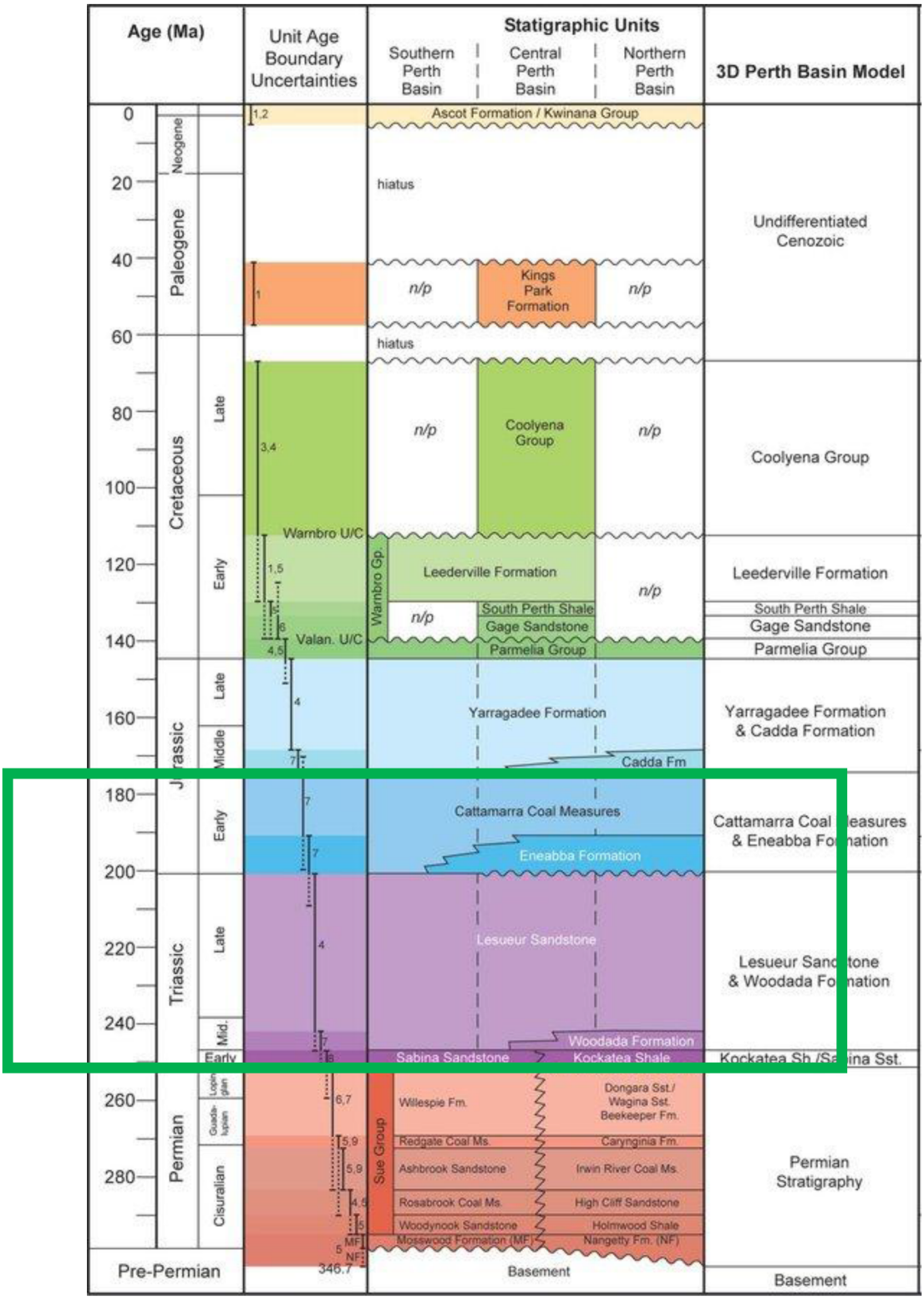



**Figure 3—Regional stratigraphy (modified from Olierook et al, 2015).**

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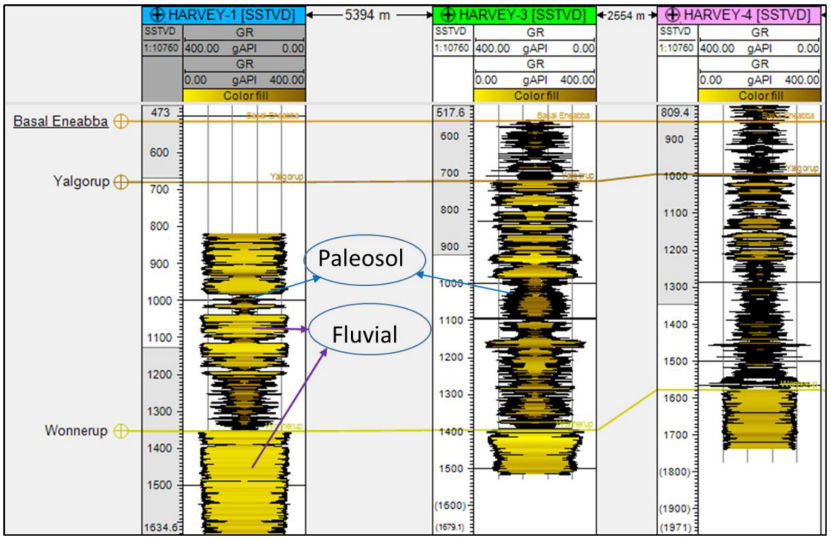



**Figure 4—Correlations between the studied formations in three of the Harvey wells for which sufficient well log data is available.**

## Methodology and available data

### Establishing Harvey Field geomechanical parameters

Determining geomechanical parameters accurately is central to any quantitative geomechanical study, and laboratory analysis of core data offers a reliable way of obtaining these properties (Dewhurst and Hennig., 2003; Lijith et al., 2019; Kolawole and Ispas., 2020). Well logs, by contrast, are recorded far more widely and give continuous coverage across formations, whereas core data is available only at limited depths. It therefore matters to establish, wherever feasible, a relationship linking well-log responses to core-derived geomechanical parameters, since such relationships allow those parameters to be extrapolated across whole wellbore sections where core is unavailable. In this study, the limited rock-mechanical data obtained from core specimens at several depths were used in this way to calibrate the well-log-derived geomechanical properties.

### Evaluation of mechanical rock properties

A first essential step in estimating in-situ stresses is measuring and calculating the rocks' elastic parameters. Young's modulus ($E$) and Poisson's ratio ($v$) can be obtained indirectly from velocity data (Khandelwal and Singh., 2009). Compressional-wave travel time (DTc) is read from sonic well logs and converted to P-wave velocity via Eq. 1, and the corresponding shear-wave travel time (DTs) is then estimated from the DTc data using Eq. 2 (Claudio Delle Piane., 2013; Pervukhina et al., 2013). Density well-log data are used with Eq. 3 to give a further velocity estimate (Gardner et al. 1974, M. Pervukhina 2017), and Eqs. 4 (Gardner et al., 1974) and 5 (Fjaer et al., 2008) were then applied to calculate well-log-derived dynamic $E$ and $v$ values for each cored interval (Wang 2000):

$$Vp = \frac{304.8}{\Delta t} \tag{1}$$

where Δt is in μs/ft

$$Vs = 0.5986\,Vp - 0.1082 \tag{2}$$

$$\rho = 1.74 V_p^{0.25} \tag{3}$$

$$E = \rho V_s^2 \frac{3V_P^2 - 4V_s^2}{V_p^2 - V_s^2} \tag{4}$$

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$$V = \frac{V_P^2 - 2V_s^2}{2\left(V_p^2 - V_s^2\right)} \tag{5}$$

in which ρ denotes formation bulk density (g/cm$^3$), Vp the compressional-wave velocity (m/s), and Vs the shear-wave velocity (m/s).

Because the correlation between the core data and the calculated results was acceptable, Eq. 6 was then used to estimate $E_{static}$ (Wang 2017).

$$E_{static} = 0.41E_{dynamic} - 1.06 \tag{6}$$

Most of the rock-strength parameters came from laboratory testing of Harvey well cores. To extend that data continuously along the well, correlations were drawn between the core results and the well logs. Several published relationships allow uniaxial compressive strength (UCS) to be estimated from elastic moduli; here, the Pervukhina relationship (Pervukhina et al., 2017) was used to calculate UCS within the clastic intervals (Eq. 7).

$$UCS = 0.05 * Vp\text{^}4.47 \tag{7}$$

where, *UCS* is expressed in MPa, and *Vp* is expressed in km/s

The friction angle ($\phi$) was also derived from the Pervukina correlation (Pervukina et al., 2017) by applying Eq. 8.

$$\phi = 0.085 * Vp\text{^}3.8 \tag{8}$$

where *Vp* is in km/s, and $\phi$ is the friction angle.

The UCS values ($C_0$) can then be combined with ($\phi$) to calculate cohesion ($S_0$) by applying Eq. 9. (Fjaer et aal., 2008)

$$S_o = C_O \frac{1 - SIN\,\emptyset}{2COS\,\emptyset} \tag{9}$$

**Estimation of initial pore pressure**

In-situ pore pressure is a key input for estimating earth stresses, which in turn feed into assessments of fault (re)activation susceptibility and wellbore stability during $CO_2$ injection. Several empirical relationships for calculating initial pore pressure appear in the literature, with the Bowers, Holbrook and Eaton approaches most commonly applied to petroleum reservoirs (Zoback 2007). Eaton's approach simplifies matters by treating compaction imbalance as the dominant mechanism behind overpressured formations (Nole et al., 2016), and on this basis Eaton (1975) proposed an empirical relationship for estimating the pore-pressure gradient from DTc well-log data (Eq. 10):

$$P_{pg} = S_g - \left(S_g - P_{ng}\right)\left(\frac{\Delta t_n}{\Delta t_o}\right)^x \tag{10}$$

where $P_{pg}$ is the pore-pressure gradient, $S_g$ is the overburden-stress gradient, $P_{ng}$ *is the* hydrostatic pore-pressure gradient, $\Delta t_n$ is the compressional sonic transit time in shales, and x is the exponent component

**In-situ stress characterization**

Both the magnitude and orientation of in-situ stresses matter when estimating how the seal rock and reservoir formation will respond to stress changes over the long term as $CO_2$ is injected. These stresses arise from a combination of gravitational and tectonic processes (Rajabi et al., 2016) and are described by three components — vertical stress ($S_V$), maximum horizontal stress ($S_H$) and minimum horizontal stress ($S_h$) — each defined by both magnitude and direction.

***In-situ stress estimation.*** $S_V$ magnitude is estimated by integrating formations density with respect to depth, as expressed by Eq. 11 (Zoback, 2007).

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$$S_V = \int_0^z \rho(z) g dz = \rho_{av} g z \tag{11}$$

where $S_V$ is overburden stress in psi, $\rho(z)$ is density in lb/ft³, $g$ is the gravitational acceleration constant, $z$ is depth in ft, and $\rho_{av}$ is the mean overburden density in lb/ft³.

A range of methods exist for predicting minimum horizontal in-situ stress, including the use of leak-off test (LOT) and extended leak-off test (XLOT) data alongside focal-mechanism analysis (Lavrov et al., 2016; Thompson et al., 2022). Here, minimum in-situ stress magnitudes were obtained from LOT and XLOT measurements, though these were not recorded continuously across the drilled Harvey wellbore sections. The poroelastic approach (Eq. 12 and Eq. 13) was therefore also applied to calculate $S_H$ and $S_h$ magnitudes.

$$S_h = \frac{v}{1-v} S_v - \frac{v}{1-v} \alpha P_p + \alpha P_p + \frac{E}{1-v^2} \varepsilon_x + \frac{vE}{1-v^2} \varepsilon_y \tag{12}$$

$$S_H = \frac{v}{1-v} S_v - \frac{v}{1-v} \alpha P_p + \alpha P_p + \frac{E}{1-v^2} \varepsilon_x + \frac{E}{1-v^2} \varepsilon_y \tag{13}$$

Where $E$ is Young's modulus, $S_v$ is vertical in-situ stress, $v$ is Poisson's ratio, $P_p$ is pore pressure, α is the Biot's coefficient, and $\varepsilon_x$ and $\varepsilon_y$ are tectonic strains.

***In situ stress orientations.*** Knowing stress directions matters for borehole-stability assessments under a range of conditions, because hydraulic tensile fractures induced by $CO_2$-injection pore-pressure increases would propagate parallel to the $S_H$ direction (Cudjoe., 2016). One practical way to identify in-situ stress azimuths is through wellbore fracture analysis of borehole image logs; here, breakout azimuth was determined by picking out the compressively stressed zones visible in the CBIL log from the Harvey-1 well.

## Boundary conditions

Because the geomechanical model extended beyond the simulated reservoir interval, the normal displacements applied for side-burden modelling were referenced to five lateral grids, and vertical displacement at the model base was likewise capped at five grids. A computed initialisation step was then used to bring these boundary conditions into mechanical equilibrium with the initial stress distribution across the study area.

## Mechanical Earth Model (MEM)

Setting up an MEM helps to better define a reservoir's geomechanical data, expressing the formations' properties as a function of depth against the stratigraphic column. Once built, an MEM can be used to inform the most suitable approaches to wellbore stability and field development. Fig. 5 summarises the MEM workflow used in this study.

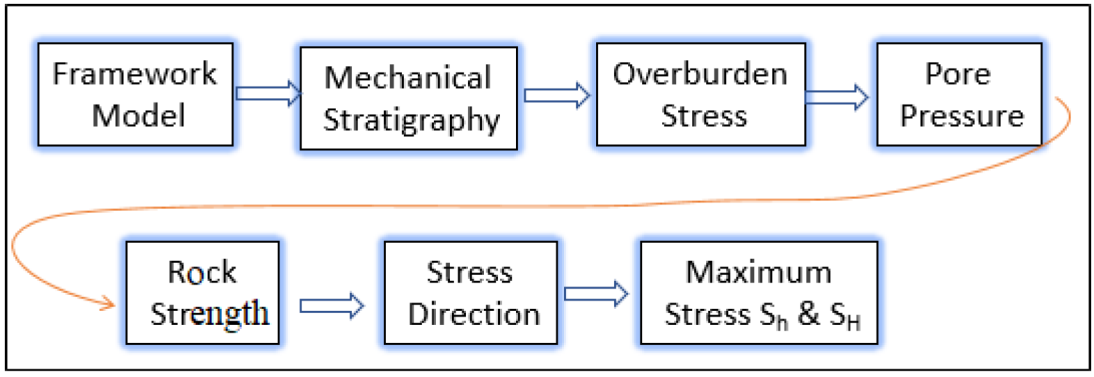



**Figure 5—Mechanical earth model (MEM) workflow adopted in this study.**

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## Coupled Geomechanical Modeling

A variety of techniques exist for coupling reservoir dynamic data with geomechanical models (Settari and Walter., 2001; Xiong et al., 2013; Zhang and Mack., 2017; McMillan et al., 2019; Jamaloei., 2021; Hawes et al., 2021), and these coupled models are useful for revealing stress-related reservoir behaviour. Fully coupled methods can resolve the complete set of geomechanical equations while accounting for hydromechanical interactions, but doing so generally forces some simplification of the hydraulic or geomechanical formulations compared with standalone, uncoupled reservoir and geomechanical models.

An alternative is a partly coupled method, built by linking standard reservoir and geomechanical simulators, in which the stress and fluid-flow equations are solved separately at each phase and data is passed back and forth between the two simulators. This tends to be a flexible approach that runs quickly in CPU terms and can readily absorb new mathematical developments as they emerge, in either the reservoir or the geomechanical simulation. The partial coupling involves several exchange steps; if each step is carried out only once, the coupling is termed ‘explicit’, whereas if it is repeated until the stress distributions and fluid flows for a given pore volume converge, it is termed "iterative".

Coupling the pore-pressure history from a dynamic reservoir simulator with the reservoir's geomechanical parameters produces what is termed a one-way partially coupled model: pore pressures from the reservoir simulation are passed through poroelastic relationships to derive stress distributions and deformation. This kind of partial coupling is an effective way to preserve the reservoir's distinct geomechanical characteristics (Settari and Walters, 2001).

This study centres on how $CO_2$ injection affects the reservoir and its cap rock. Where the pore-pressure profile stays comparatively low, deformation is unlikely to noticeably alter reservoir properties such as porosity, so a coupled geomechanical model is generally unnecessary. Because CCS reservoirs receive $CO_2$ in multiple stages, however, it is worth applying a one-way coupling approach to gauge how susceptible the existing faults are to reactivation as injection continues over time. Thermal effects arising from injecting cool $CO_2$ into the warmer reservoir have been left out of this study. This omission is deliberate rather than incidental. The pore-pressure and saturation histories used to drive the geomechanical model were produced by an isothermal dynamic simulation, so no reservoir temperature field was available to impose as a thermal load, and because the Harvey site is still pre-injection, the injection temperature, the rate-dependent Joule-Thomson cooling and the extent of the resulting near-well cooling front would all have to be assumed rather than measured. Cooling of this kind contracts the rock skeleton and reduces the effective horizontal stresses within the cooled volume, which lowers the margin against tensile and shear failure close to the injectors and at the base of the seal (Gor et al., 2013; Selvadurai and Suvorov, 2016; Mathur et al., 2024). That volume is confined to the immediate vicinity of the injection wells, however, whereas the faults assessed here lie several hundred metres to kilometres away, where the stress perturbation is governed by pore-pressure change rather than by temperature. The isothermal treatment is therefore considered adequate for the field-scale fault-reactivation question addressed in this paper, and the results reported below should be read as the pressure-driven response; a fully coupled thermo-hydro-mechanical analysis is required before any statement can be made about near-wellbore and near-seal integrity within the cooled region, and forms the subject of ongoing work.

The 3D MEM of the Harvey reservoir's stress field was built using the finite-element method (FEM) in Visage. FEM subdivides the geological formation into numerous finite elements (FEs) joined at shared nodes; each FE carries its own rock mechanical properties, which are progressively updated as loads are applied across the reservoir grid, and the resulting property changes are propagated through the model to derive the strain associated with the evolving stress magnitude at each FE.

Accounting for tectonic stresses in the modelling workflow involves three steps: identifying the relevant tectonic events and features, including faults and the structure and stratification of the reservoir and cap rock; mapping the lateral distribution of formation elastic moduli across the study area; and defining the

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initial stress regime together with suitable boundary-condition assumptions. Fig. 6 highlights (within the dashed-blue polygon) the geomechanical modelling workflow followed in this study.

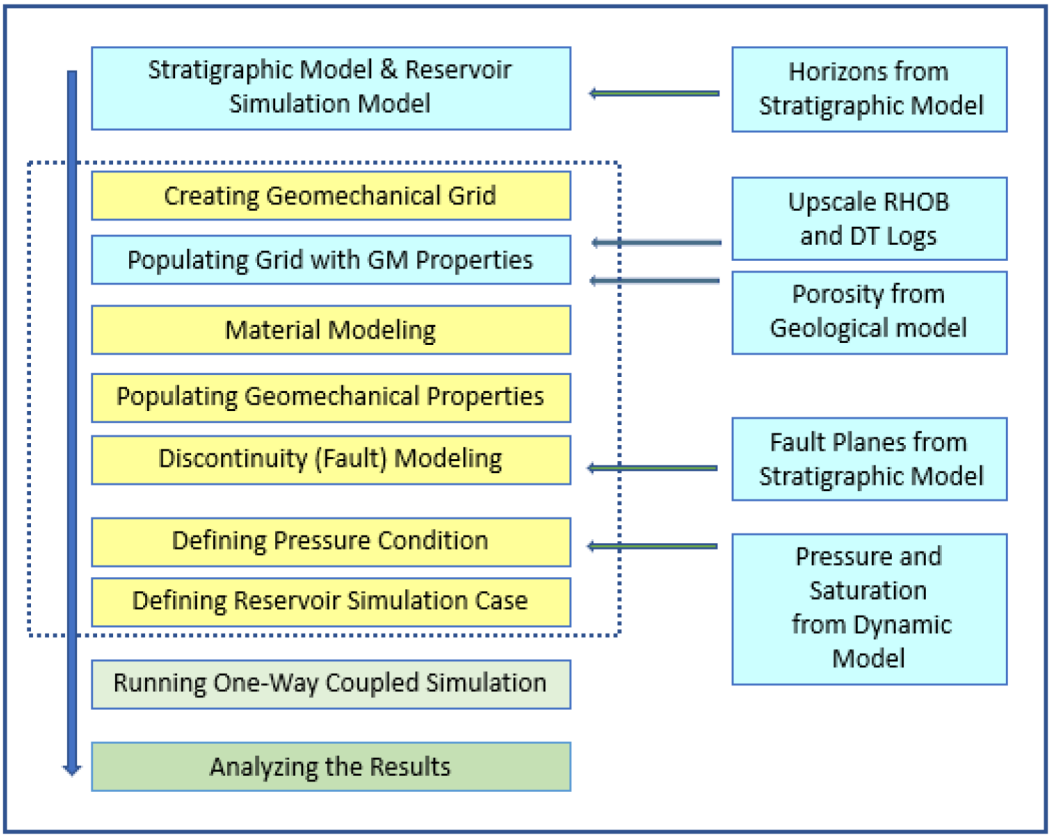



**Figure 6—Coupled Reservoir-Geomechanical model workflow highlighted in the dashed blue area.**

## Result and Discussion

### 3D Mechanical Earth Model (MEM)

Building an MEM makes fuller use of the available geomechanical data, mapping the formations' mechanical properties across the whole study area. A properly calibrated MEM can then be used to identify the most suitable approaches for safe $CO_2$ injection and field development, with the 3D modelling outputs providing field-wide property distributions drawn from combined seismic and well data.

***Geometry.*** Defining the reservoir boundary conditions is typically the most time-consuming part of building a geomechanical FEM model. Formation surfaces and fault polygons are usually derived from seismic interpretation, which is well suited to constraining stratigraphic and structural models for the area of interest, and an upscaled geological grid can usefully be carried into the geomechanical model. Even so, the FE grid still has to satisfy certain requirements, most importantly that it fully covers the reservoir boundaries (Fig. 7). Sideburdens — limiting boundaries built parallel to the reservoir grid direction — were added around the studied area, with a lithostatic load representing the sedimentary overburden applied as the top boundary condition; the cap rock is treated as part of the overburden and so is excluded from the modelled volume. The underburden was extended below the constructed grids to improve the model's aspect ratio and give a more realistic transfer of stress from the boundaries into the reservoir and overburden during stress modelling.

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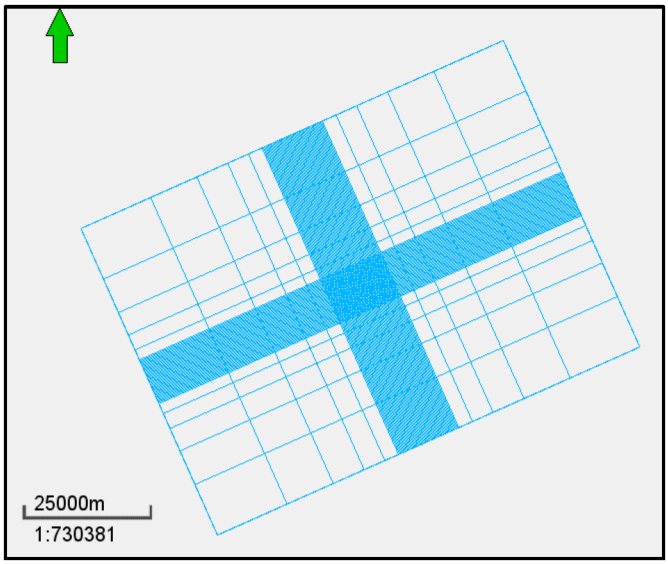



**Figure 7—Geomechanical gridding of the study area. Sideburden, overburden, and underburden cells (large blue cross) considered to remove the edge effects.**

Five extra cells were added around the sideburdens, growing geometrically by a factor of 1.5 towards the model edges (Fig. 7), and stiff 50 m-thick 'plates' were placed at the lateral margins of the grid. For simplicity, these constraining plates are treated as uniform, homogeneous material volumes; they could be modelled in finer detail, including all horizons and layers, once sufficient structural and material data becomes available.

### *Mechanical property variations.*

### *Elastic Properties*

Rock mechanical parameters are essential to establish accurate predictions of UCS and earth stresses, which can later be refined and adjusted as additional information becomes available. Fig. 8 A to F illustrate the distribution of the velocity, density and mechanical properties throughout the studied area. $E_{stat}$ and $\nu_{stat}$ differ vertically in the reservoir. $V_p$, $V_s$, $\rho$, $\nu$, $G$, and $E$ average values are 3 km/s, 1.75 km/s, 2.3 g/cm, 0.22, 10 GPa, and 5 GPa for Yalgroup Formation, respectively. For Wonnerup Formation these average values are 4 km/s, 2.5 km/s, 2.5 g/cm, 0.18, 25 GPa, and 18 GPa, respectively.

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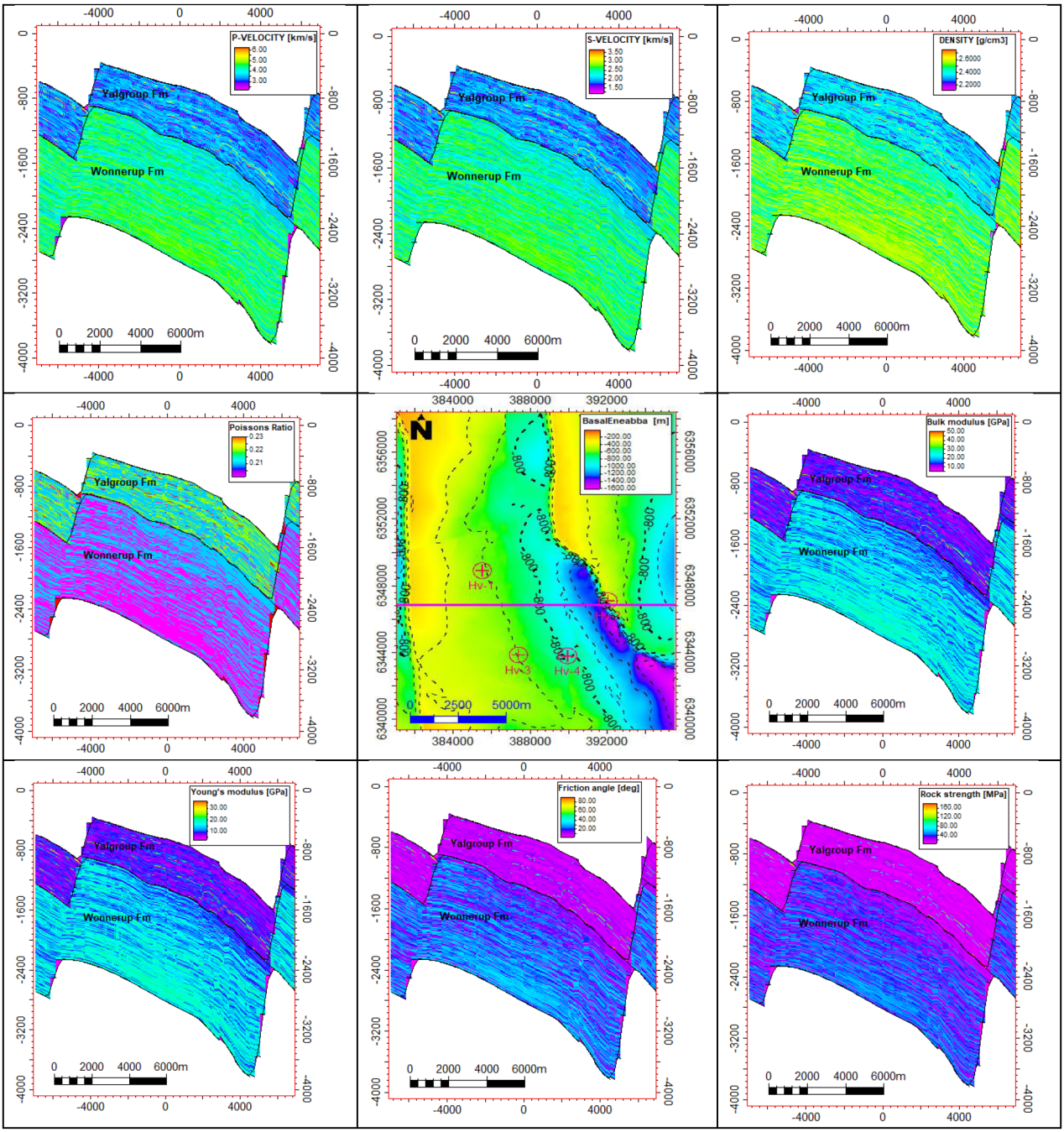



**Figure 8—P & S velocities, density, elastic, rock strength (UCS), and friction angle parameters for Yalgorup and Wonnerup reservoirs in the studied area.**

### *Rock Strength*

How well a formation withstands a given earth-stress regime depends on its rock-strength parameters, commonly expressed as unconfined compressive strength (UCS). UCS was estimated from well-log data using the Pervukhina correlation (Eq. 7) (Pervukhina et al., 2017), which draws on Young's modulus, shear modulus, porosity and other rock properties. Fig. 8 shows average UCS values of 20 MPa for the Yalgorup Formation and 40 MPa for the Wonnerup Formation. At smaller scales, nanoindentation-based fracture-toughness measurements offer a complementary means of characterising rock-strength heterogeneity where core material is limited (Liu et al., 2023).

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For friction angle ($\phi$) determinations, high values of this parameter suggested by rock-mechanical tests were ignored. Instead, the friction angle was calculated using available $V_p$ data. The results reveal the average $\phi$ values of about 10 deg and 40 deg for Yalgorup and Wonnerup Formations, respectively (Fig. 8). In addition, the average vertical permeability to horizontal permeability ratio (kv/kh) is assumed to be 0.75. Fig. 9 shows the distribution of porosity and permeability properties for the studied reservoir.

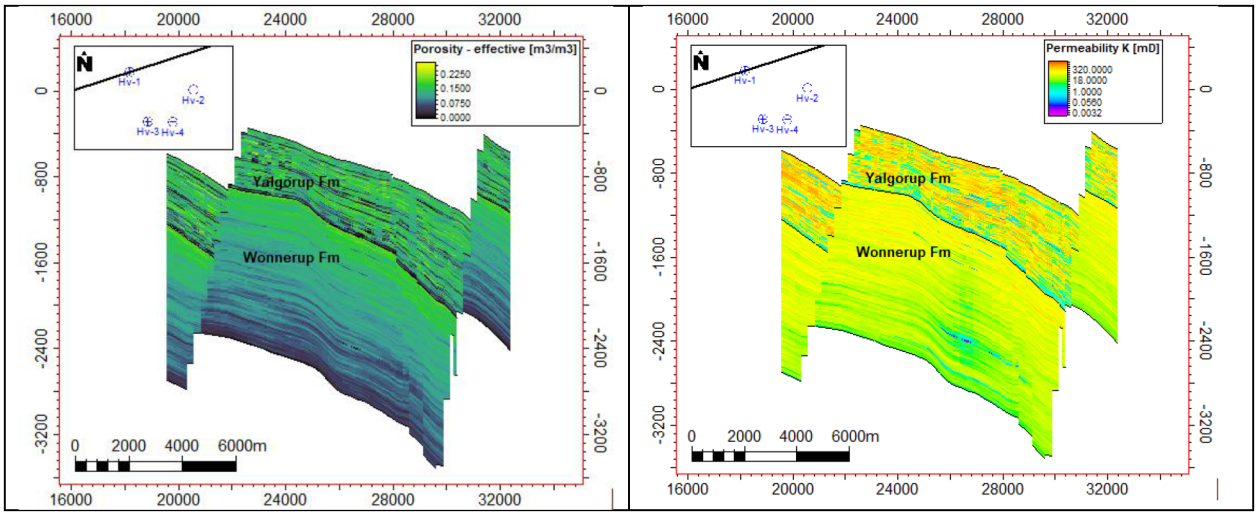



**Figure 9—Porosity and permeability parameters for Yalgorup and Wonnerup reservoirs in the studied area.**

## *Pore Pressure*

A number of stress-dependent methods exist for deriving pore pressure from well-log data. Here, both the Eaton and Bowers methods were applied to estimate pore-pressure profiles within the clay-rich overburden intervals. The Eaton method was ultimately adopted for the non-reservoir intervals, as it matched the observed drilling-fluid densities and drilling events more closely. Fig. 10 presents the resulting pore-pressure distribution across the studied formations: Fig. 10A shows the modelled vertical profile at the start of $CO_2$ injection, and Fig. 10B shows the pore-pressure distribution in plan view across a depth section at 3055 m.

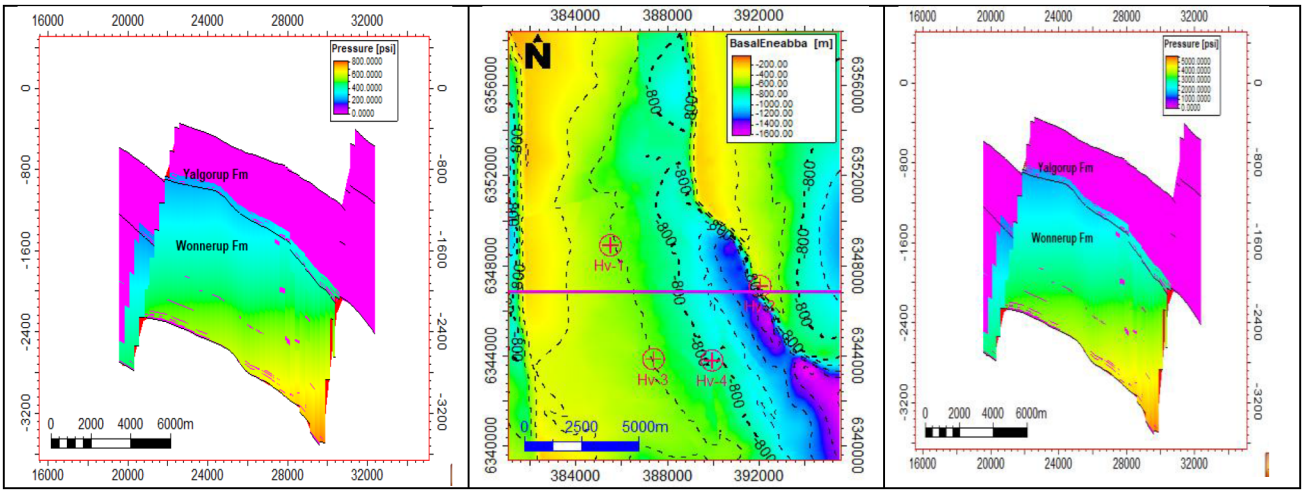



**Figure 10—Distribution of pore pressure at two-time steps in the simulation. The time step of the left cross-section is 2020. The time step of the right cross-section is 2050.. as it shown, the pore pressure value range for the 2020 timestep is up to 500 psi, and for 2050 timestep is up to 5000 psi.**

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## Discontinuity Modeling

Both minor and major faults were mapped and added into the model at this stage (Fig. 11). Comparing different models requires a common basis, so laboratory-derived correlations were used for the initial models; where literature-based correlations specific to the Perth Basin gave a closer match for certain parameters, these were substituted instead (Zhang et al. 2015), as summarised in Table 1.

**Table 1—Material properties considered by Zhang et al 2015 for the South West Hub.**

| Unit | Density (Kg/m³) | Young's Modulus (GPa) | Poisson's ratio | Cohesion (MPa) | Tensile strength (MPa) | UCS (MPa) | Porosity | Friction (◦) | Dilation angle (◦) |
|---|---|---|---|---|---|---|---|---|---|
| Leederville Fm | 2265 | 14 | 0.26 | 7 | 3.5 | 20.8 | 0.12 | 22 | 2 |
| Guildford Fm | 2301 | 14 | 0.26 | 7 | 3.5 | 20.8 | 0.12 | 22 | 2 |
| Eneabba Fm | 2352 | 14 | 0.26 | 7 | 3.5 | 20.8 | 0.13 | 22 | 2 |
| Basal Eneabba shale | 2400 | 2 | 0.35 | 2.75 | 1.375 | 8.2 | 0.05 | 22 | 2 |
| Yalgorup member | 2260 | 7.9 | 0.23 | 5.28 | 2.64 | 12.9 | 0.12 | 11.25 | 2 |
| Wonnerup member | 2269 | 20.3 | 0 295 | 14.1 | 7.05 | 50.7 | 0.13 | 31.87 | 2 |
| Sabina sandstone | 2269 | 20.3 | 0 295 | 14.1 | 7.05 | 50.7 | 0.13 | 31.87 | 2 |
| Willespie Fm | 2265 | 14 | 0 26 | 7 | 3.5 | 20.8 | 0.12 | 22 | 2 |
| Weak fault | 2280 | 2 | 0 15 | 2 | 1 | 5.2 | HRP * | 15 | 3 |
| Strong fault | 2280 | 40 | 0.3 | 28 | 14 | 97.0 | HRP* | 30 | 3 |

Host rock porosity

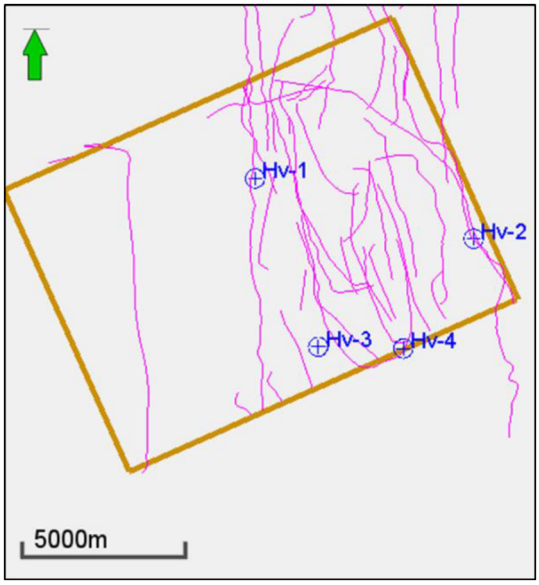



**Figure 11—Existing faults (shown in pink) identified by discontinuity modeling.**

## Boundary Condition

***Horizontal Stress Direction.*** Establishing the horizontal stress orientation is one of the first tasks in stress characterisation and modelling, and stress-direction analysis is usually a central objective of geomechanical studies more broadly. Knowing both the orientation and magnitude of the stresses helps to minimise the risks of $CO_2$ injection and to optimise injection strategy. Because stress concentration is greatest along the minimum horizontal stress ($S_h$) direction, wellbore breakouts commonly extend from the borehole in that

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orientation, so breakout azimuth analysis in a vertical well can be used to identify $S_h$. Here, CBIL data from the Harvey-1 well was interpreted for this purpose, giving a breakout direction of roughly zero degrees across the study area (Fig. 12).

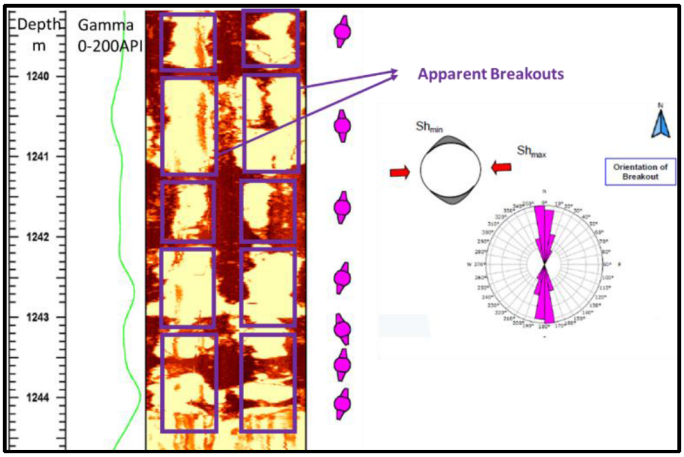



**Figure 12—Direction of identified break-outs interpreted using CBIL data from Harvey 1 well.**

Where the stress regime is tensional, induced fractures tend to run along a borehole parallel to the maximum principal stress ($S_H$) azimuth, so the induced fracture azimuth indicates the $S_H$ direction in a vertical well. In this study, maximum horizontal stress was taken to lie perpendicular to the minimum horizontal stress direction, at an orientation of 90 degrees, as shown for the field location in Figure 13.

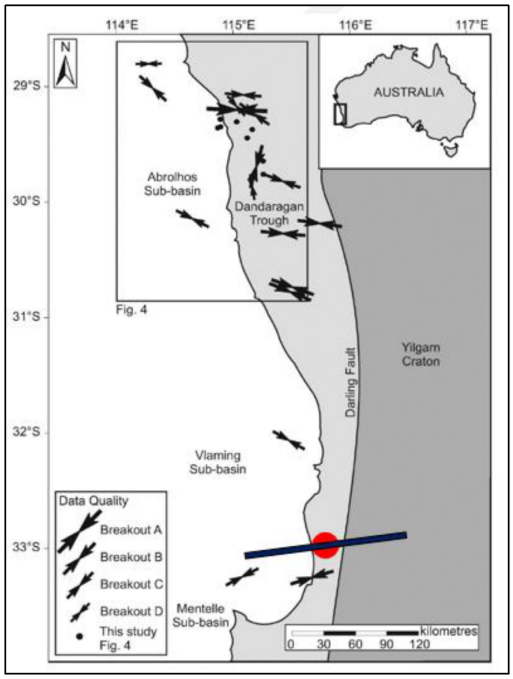



**Figure 13—Maximum horizontal stress orientations for the Perth Basin. (modified after Hillis and Reynolds., 2000). The thick black line indicates the direction SH from the wellbore breakout analysis of Harvey-1.**

***Vertical and Horizontal Stress Magnitudes.*** Because in-situ stress magnitudes are hard to measure directly, they are instead estimated from other measurable variables such as deformation (strain) and pressure. Even so, various factors constrain both the magnitude and orientation of the principal stresses ($S_v$, $S_h$ and $S_H$) within a formation. Since formation density is readily available from most well-log suites, bulk density logs

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(ρb) were used here to determine $S_V$, integrating (ρb) from the surface to total well depth via Eq. 11; the resulting $S_V$ profile for the Harvey-1 well is shown in the first track of Fig. 14.

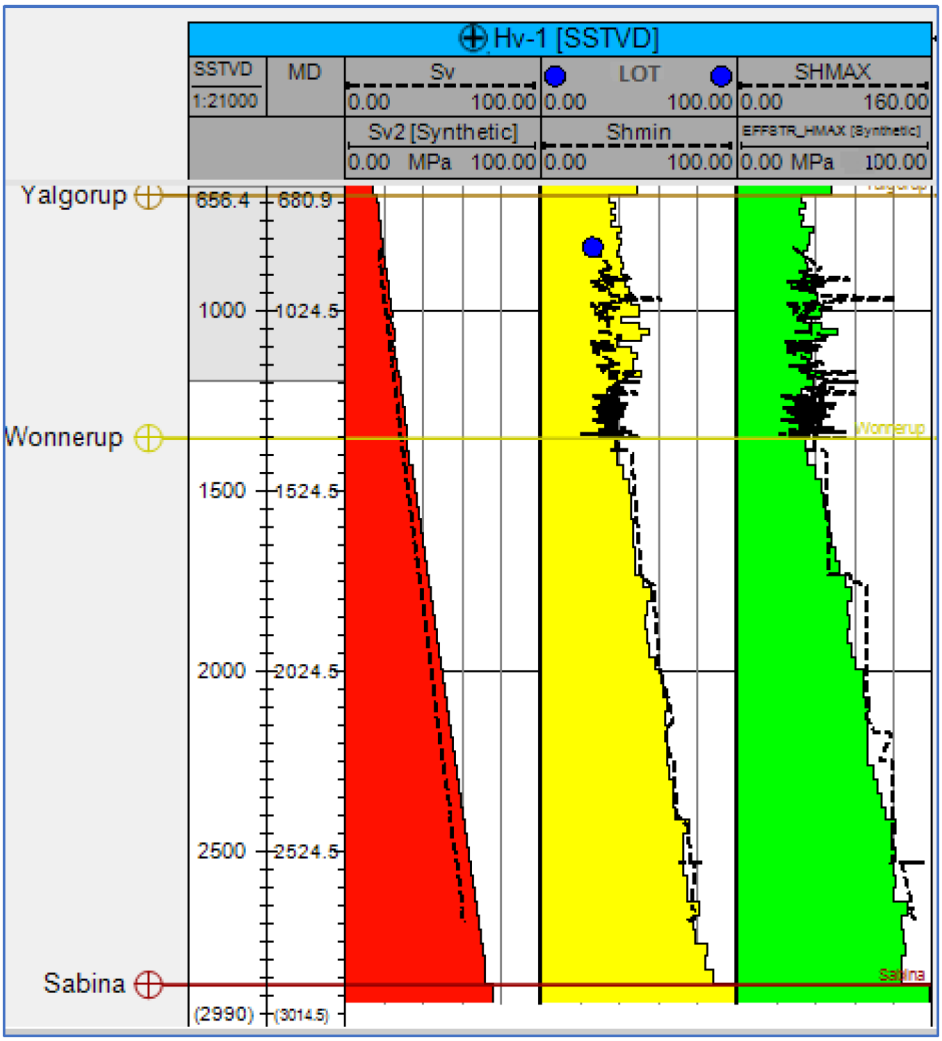



**Figure 14—Vertical ($S_v$) and both horizontal stresses ($S_H$ and $S_h$) modeling results determined for the Harvey-1 well section. The estimated value for Shmin is well-matched with LOT result (second track).**

Poroelastic horizontal strain modelling depends on obtaining reliable tectonic strain values. The two horizontal tectonic strain factors, εx and εy, which can be either compressional or extensional, serve as useful calibration parameters that can be tuned to improve the fit of the stress estimates obtained from $S_h$ calculations.

An initial estimate of the stress field is needed as a starting point from which the likely range of stress magnitudes, particularly $S_H$, can be assessed. These initial estimates were then refined by adjusting εx and εy until the predictions matched all of the regionally available stress-field indicators (Fig. 13); the best-fitting values were 0.1 for εx and 0.9 for εy.

The model outputs stress magnitudes in any direction across the study area; for the Harvey-1 well, Fig. 14 shows $S_v$ ranging from 17 to 60 MPa, $S_H$ from 22 to 75 MPa, and $S_h$ from 21 to 72 MPa.

The simulated stress regime also helps further characterise the geomechanics of the modelled subsurface formations, with empirical stress ratios (Eq. 12 and Eq. 13) used to represent potential stress anisotropy across the study area.

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Boundary conditions have a strong bearing on finite-element modelling, imposing regional constraints on model behaviour. These are applied to the whole model, including the sideburdens, either as loads (tectonic stresses) or as deformations (tectonic strains), and are translated into the finite elements at the start of every simulation run. For this study, the Harvey area's prevailing stress conditions were established through image-log interpretation following standard procedures (King et al. 2008). Combining these boundary-condition assumptions with the integrated model data gives an average $S_H$ azimuth of 90°, consistent with regional expectations (Fig. 13).

The horizontal stress magnitudes carry their own uncertainty, so average model-generated stress values were used to derive estimated stress gradients. At 1500 m depth, $S_h$ is 32 MPa and $S_H$ is 37 MPa, giving an $S_H/S_h$ ratio of about 1.125. The horizontal stress range for the study area also needs to be consistent with the LOT results from the Harvey well data, summarised in Table 2. Tectonic strains were introduced into the model in the same way, using the approximate strain values shown in Table 3, which were likewise drawn from LOT test results obtained across the Harvey field.

**Table 2—Summary of Leak of Test results for the South West Hub**

| Well Name | Leak-off test | Depth (m) | Formation | Pump pressure (psi) | LOT (MPa) |
|---|---|---|---|---|---|
| Hv#2 | Successful | 209 | Eneabba Formation | 375 | 7.4 |
| Hv#3 | Successful | 591.4 | Eneabba Formation | 640 | 11.5 |
| Hv#3 | Formation Integrity Test | 1467.4 | Wonnerup Member | 1725.5 | 29.4 |
| Hv#4 | Successful | 267 | Eneabba Formation | 214 | 4.2 |
| Hv#1 | Successful | 848 | Yalgroup Formation | NA | 26.23 |

**Table 3—Strain, $\varepsilon_x$, in the direction of a minimal horizontal stress and the strain ratio K. These values are derived from the LOT test results, observed throughout the South West Hub.**

| Case | $S_h$ (MPa) | $\varepsilon_x$ | $K= \varepsilon y/\varepsilon_x$ |
|---|---|---|---|
| 1 | 29.4 at 1467 m | 0.50 | 1.7 |
| 2 | 7.4 at 400 m | 0.02 | 50 |
| 3 | 26 at 800 m | 0.90 | 1.2 |

Repeated hydrostatic simulations showed that a $S_h$ gradient of 0.18 MPa gave the closest match between the FEM-derived horizontal stress values and the LOT results, as shown in Fig. 14

## Coupling of Geomechanical and Dynamic Models

The conceptual development plan for the Harvey area calls for injecting 800,000 tonnes of $CO_2$ per year over a 30-year period, after which the wells are shut in and the $CO_2$ dissipates through the aquifer. The reservoir simulations examine a range of injection scenarios and subsurface uncertainties to build confidence that the $CO_2$ plume will remain below 800 mTVDss and within the storage complex for 1000 years.

A "Reference Case" was defined for the injectivity studies, using a bottom-hole-pressure constraint of 360 bars at 2948 m [Pore Pressure + 0.9×69 bar]. The modelled well configuration comprises nine wells arranged in a staggered line-drive pattern (Figure 15), all completed in the lowest 250 metres of the Wonnerup formation, through which $CO_2$ is injected into the Wonnerup reservoir. Outputs from this reference case are fed into the geomechanical model.

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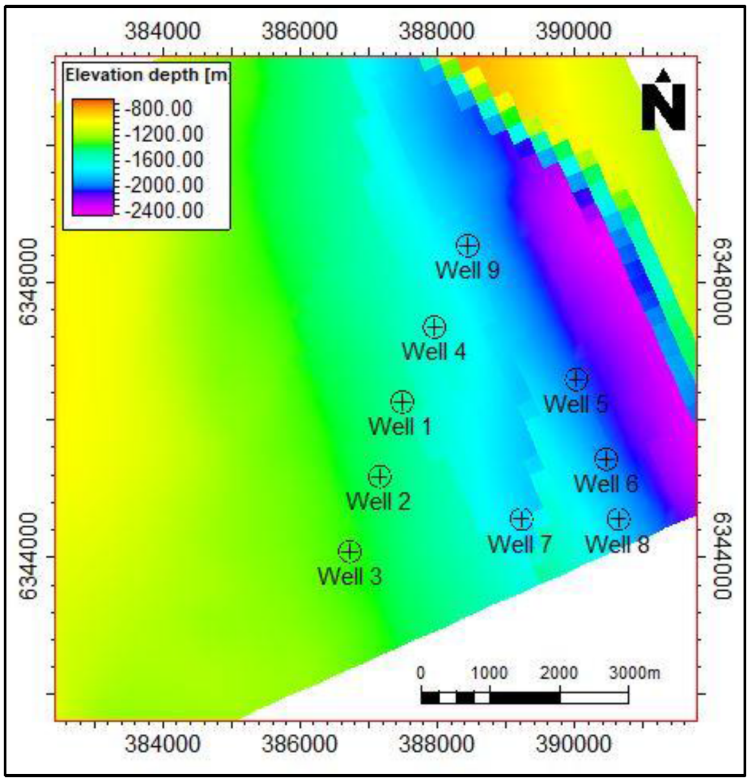



**Figure 15—Location of injection wells in relation to the simulated rock volume.**

Pore-pressure changes were calculated from a dynamic reservoir simulation projecting $CO_2$ injection from 2021 through to 2035. Five pore-pressure and water-saturation profiles were evaluated and coupled to the geomechanical model, which was then re-run repeatedly to refine the stress-field outputs for the reservoir volume and calibrate them against the available LOT test data.

## Simulation validation

Repeated iteration eventually brought the simulation model to convergence on grid characteristics consistent with the recorded and calculated geomechanical properties and boundary-condition assumptions. It is this final converged iteration of the GM, rather than the earlier ones, that is used to calibrate the model by comparing $S_h$ values at the LOT test locations. Once the GM's initial conditions had been adjusted to these calibrated values, $CO_2$ injection into the studied Harvey reservoir volume was analysed in detail. Hydrostatic conditions were assumed for the overburden pore-pressure calculations, allowing the model to be run and the predicted pore pressure calibrated against the in-situ stress results; each simulation run produces a substantial volume of calculated variable data.

## Faults Analysis

Before the geomechanical model (GM) was built, the available data — formation properties, horizons and fault surfaces — were reviewed within the upscaled geology model, which uses a structural grid with a larger cell area of 250×250 m. The GM can then forecast the reservoir stress field during $CO_2$ injection, project overburden movement as injection proceeds, and assess the potential for existing faults penetrating the studied formations to destabilise or mobilise as injection continues. This GM was built from the upscaled 3D geological and reservoir models, with its main components shown in Fig. 15.

Changes in stress state during $CO_2$ injection could potentially reactivate existing faults and, in turn, cause caprock leakage, so mapping and investigating faults is essential to assessing reservoir and cap-rock stability throughout the injection period. More than 30 faults were mapped across the field (Fig. 11), predominantly

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oriented N-S in keeping with regional fault trends. On a Mohr-Coulomb chart, stress states are expressed in terms of $S_H$, $S_h$ and $S_v$ — the maximum, intermediate and minimum principal stresses, respectively — and fault stability is expected to shift as this stress state changes (Wu and Hasegawa., 1996; Ganguli and Sen., 2020; Mortazavi and Maratov, 2024; Mathur et al., 2024). The outer Mohr circle has radius $(S_H\text{-}S_v)/2$ and is centred at $(S_H+S_v)/2$, and under the Mohr-Coulomb failure criterion, rock failure occurs once this circle touches the failure envelope (Noel et al., 2019; Van den Bogert and Van Eijs., 2020). As pore pressure rises with continued $CO_2$ injection, the reservoir's effective stress magnitudes shift and the calculated Mohr circle moves leftward, directly reflecting the change in the reservoir's stress state (Leclere and Fabbri., 2013; Hedayatzadeh et al., 2022) (Fig. 16).

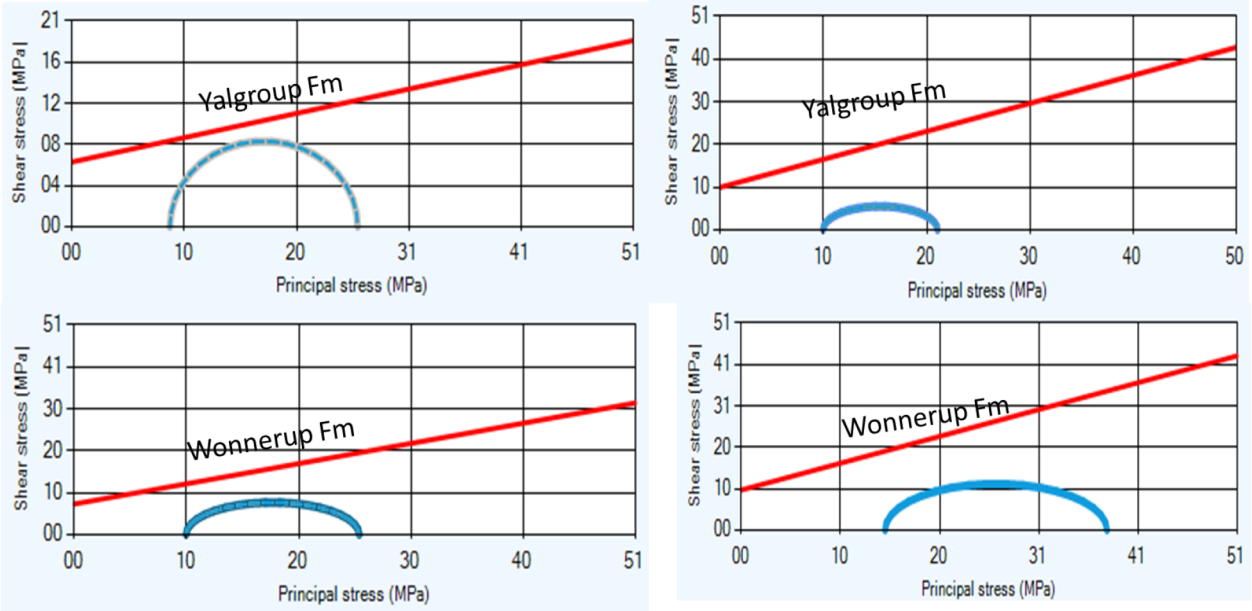



**Figure 16—Mohr Circle plots for four timesteps, including cell 40-29-49 at 2020 timestep, cell 40-39-26 at 2030 timestep, cell 50-36-49 at 2035 timestep, and cell 50-31-69 at 3050 timestep. The Mohr circle moves to the left and changes the stress magnitudes due to the increasing the pore pressure as $CO_2$ injection progresses, causing alteration in the stress state in the reservoir.**

***Stress Charting and Evidence of Rock Failure.*** The stress-charting technique offers a way to interrogate each simulation iteration's predictions, establishing a stress path that links changes in stress to changes in pore pressure. For every cell in the simulation model, the Mohr circle can be plotted against the failure envelope (Fig. 16): the area enclosed by the blue semi-circle in Fig. 16 tends to expand as pore pressure rises, and would eventually breach the failure envelope, signalling possible formation failure, including fault reactivation. As Fig. 16 shows, however, no such breach of the failure envelope occurred at any timestep or cell location within the Yalgorup or Wonnerup formations over the simulated injection period, indicating no possibility of failure in either formation. The stress-charting process also yields the principal stress magnitudes ($S_V$, $S_H$, $S_h$) and the $S_h$ and $S_H$ orientations for every simulation grid cell, providing useful quality-control information for assessing model performance and supporting the derivation of stress–strain relationships. This result likewise holds for the isothermal case; the Mohr circles shown do not carry the additional thermal contribution to effective stress that would arise in the cooled region near the injection wells.

***Reservoir Yield Criteria and Evidence of Fault Reactivation.*** The yield mode reached in each part of the simulated reservoir volume is colour-coded in the legend of Fig. 17 (right), using YIELDMOD values at the 2050 timestep: code -2 (pink) denotes no failure, code -1 (blue) tension, code 0 (green) shear, and code 2 (red) cap failure. None of the simulation cases evaluated showed failure within the modelled Wonnerup or

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Yalgorup formation layers. Because the model is isothermal, these yield results describe the pressure-driven response only. Within the cooled volume around the injection wells, thermal contraction would further reduce the effective horizontal stresses, so the failure tendency in that near-well region is likely to be under-predicted here.

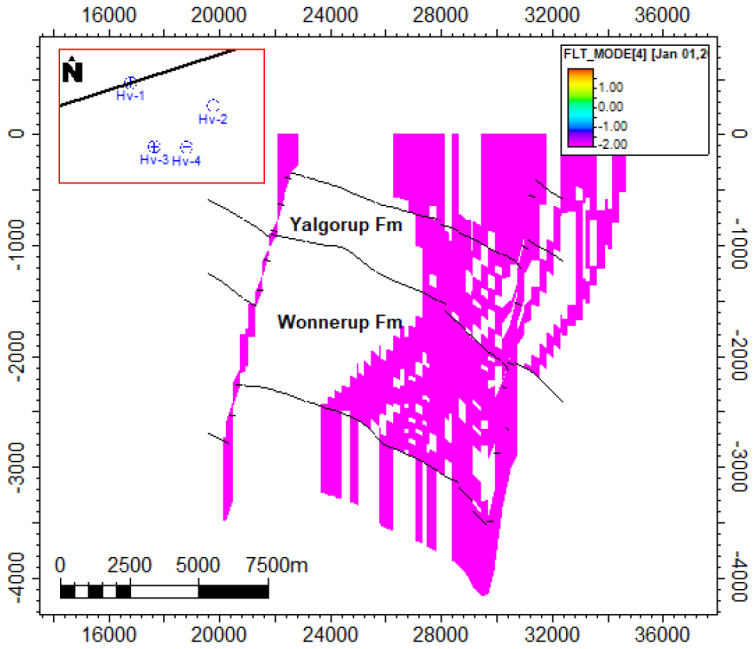



**Figure 17—YIELDMOD values at 2050 timestep in the simulated reservoir volume. Pink colour (code number -2 on the FLT_Mode scale) indicates no rock failure under applied pressure at 2050 timestep through the selected intersection. The above-left map shows the azimuth and the place of the intersection.**

## Conclusions

Bringing together static geological, dynamic reservoir and geomechanical models within a 4D finite-element framework gives a meaningful picture of the stress state of the South-West Hub (SWH) Carbon Capture and Storage (CCS) reservoir in Western Australia, based on data from the Harvey wellbore. Multiple 1D and 3D iterations were carried out during modelling to calibrate the Sh/SH values, validate the wellbore data and account for fault activation within the reservoir formation.

The model estimated $S_v$ to range between 17 and 60 MPa, $S_H$ between 22 and 75 MPa, and $S_h$ between 21 and 72 MPa. More than thirty faults were mapped across the field, predominantly trending N-S, consistent with the dominant regional fault orientation.

After optimising the coupled model, the reservoir's in-situ stress system was modelled in stages through a series of optimisation runs. Overall, the results point to no significant risk of strain or fault failure in the Harvey reservoir under the predicted pressure changes.

The area representing principal stress tends to grow as pore pressure rises, potentially reaching the rock failure envelope and increasing the chance of formation failure, including fault reactivation. Nonetheless, the Mohr-circle results from the model simulation show no possibility of failure occurring in either the Yalgorup or Wonnerup formations, at any timestep through to 2050 and at any modelled cell location. The fault yield mode parameters generated at each modelled timestep likewise showed no failure anywhere in the reservoir section of the Harvey field.

These conclusions apply to the isothermal, pressure-driven response of the reservoir. Thermal stresses generated by injecting $CO_2$ at a temperature below that of the formation are not represented in the model, and such stresses can be significant close to the injection wells and at the base of the seal. The finding that no fault reactivation is expected is therefore robust at the field scale, where pore-pressure change dominates the

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stress path, but it should not be read as a statement about near-wellbore or caprock integrity within the cooled region. A fully coupled thermo-hydro-mechanical analysis, informed by measured injection temperatures, is recommended as the next step in assessing the site.

**Declaration of interests**

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

□ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

## References


Abdollahi, R., Hosseinzadeh, S., Movassagh, A., Kasperczyk, D., & Haghighi, M. (2024). Imaging pressure distribution in geological reservoirs from surface deformation data. *Sustainability*, **16**(17), 7710.

Amiri, M., Lashkaripour, G. R., Ghabezloo, S., Moghaddas, N. H., & Tajareh, M. H. (2019). Mechanical earth modeling and fault reactivation analysis for CO2-enhanced oil recovery in Gachsaran oil field, south-west of Iran. *Environmental Earth Sciences*, **78**(4), 1–22.

Alabdulkarem, A., Hwang, Y., & Radermacher, R. (2012). Development of CO2 liquefaction cycles for CO2 sequestration. *Applied Thermal Engineering*, **33**, 144–156.

Al-Ameri, W. A., Abdulraheem, A., & Mahmoud, M. (2016). Long-term effects of CO2 sequestration on rock mechanical properties. *Journal of Energy Resources Technology*, **138**(1).

Anggara, F., Sasaki, K., & Sugai, Y. (2016). The correlation between coal swelling and permeability during CO2 sequestration: A case study using Kushiro low rank coals. *International Journal of Coal Geology*, **166**, 62–70.

Bachu, S. (2000). Sequestration of CO2 in geological media: criteria and approach for site selection in response to climate change. *Energy conversion and management*, **41**(9), 953–970.

Bachu, S., & Adams, J. J. (2003). Sequestration of CO2 in geological media in response to climate change: capacity of deep saline aquifers to sequester CO2 in solution. *Energy Conversion and management*, **44**(20), 3151–3175.

Bhattacharyya, S. S., Leite, F. F. G. D., Adeyemi, M. A., Sarker, A. J., Cambareri, S., Faverin, C., … & Parra-Saldívar, R. (2021). A Paradigm Shift to CO2 Sequestration to Manage Global Warming–With the Emphasis on Developing Countries. *Science of The Total Environment*, 148169.

Birkholzer, J., Cihan, A., & Bandilla, K. (2014). A tiered area-of-review framework for geologic carbon sequestration. *Greenhouse Gases: Science and Technology*, **4**(1), 20–35.

Borissova, I., Bradshaw, B., Nicholson, C., Payne, D., & Struckmeyer, H. (2010). Mentelle Basin? tectonic evolution controlled by of the combined extensional history of the Southwestern and Southern Australian margins. *ASEG Extended Abstracts*, **2010**(1), 1–4.

Bowers, G. L. (1995). Pore pressure estimation from velocity data: Accounting for overpressure mechanisms besides undercompaction. *SPE Drilling & Completion*, **10**(02), 89–95.

Cai, F., Ding, L., Yao, W., Laskowski, A. K., Xu, Q., Zhang, J. E., & Sein, K. (2017). Provenance and tectonic evolution of Lower Paleozoic–Upper Mesozoic strata from Sibumasu terrane, Myanmar. *Gondwana Research*, **41**, 325–336.

Chen, B., Harp, D. R., Lu, Z., & Pawar, R. J. (2020). Reducing uncertainty in geologic CO2 sequestration risk assessment by assimilating monitoring data. *International Journal of Greenhouse Gas Control*, **94**, 102926.

Crostella, A., & Backhouse, J. (2000). *Geology and petroleum exploration of the central and southern Perth Basin, Western Australia (No. 57).* Perth, WA: Geological Survey of Western Australia.

Cudjoe, S., Vinassa, M., Gomes, J. H. B., & Barati, R. G. (2016). A comprehensive approach to sweet-spot mapping for hydraulic fracturing and CO2 huff-n-puff injection in Chattanooga shale formation. *Journal of Natural Gas Science and Engineering*, **33**, 1201–1218.

Dewhurst, D. N., & Hennig, A. L. (2003). Geomechanical properties related to top seal leakage in the Carnarvon Basin, Northwest Shelf, Australia. *Petroleum Geoscience*, **9**(3), 255–263.

Dillinger, A., & Esteban, L. (2014). Experimental evaluation of reservoir quality in Mesozoic formations of the Perth Basin (Western Australia) by using a laboratory low field nuclear magnetic resonance. *Marine and Petroleum Geology*, **57**, 455–469.

Doughty, C., Freifeld, B. M., & Trautz, R. C. (2008). Site characterization for CO 2 geologic storage and vice versa: the Frio brine pilot, Texas, USA as a case study. *Environmental Geology*, **54**(8), 1635–1656.

Espinoza, D. N., & Santamarina, J. C. (2017). CO2 breakthrough—Caprock sealing efficiency and integrity for carbon geological storage. *International Journal of Greenhouse Gas Control*, **66**, 218–229.


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Frash, L. P., Carey, J. W., Ickes, T., & Viswanathan, H. S. (2017). Caprock integrity susceptibility to permeable fracture creation. *International Journal of Greenhouse Gas Control*, **64**, 60–72.

Ferdinando, D. D., & Longley, I. (2015). Permian play mapping in the northern Perth Basin. *Petroleum in Western Australia*, 17–26.

Hall, L., Borissova, I., Grosjean, E., Southby, C., Owens, R., Bernardel, G., & Mitchell, C. (2017). *Petroleum Prospectivity of the Houtman Sub-Basin, Offshore Perth Basin, Australia.* In AAPG/SEG International Conference and Exhibition, London (pp. 15–18).

Harris, L. B. (1994). Structural and tectonic synthesis for the Perth Basin, Western Australia. *Journal of Petroleum Geology*, **17**(2), 129–156.

Harp, D. R., Stauffer, P. H., O'Malley, D., Jiao, Z., Egenolf, E. P., Miller, T. A., … & Pawar, R. (2017). Development of robust pressure management strategies for geologic CO2 sequestration. *International Journal of Greenhouse Gas Control*, **64**, 43–59.

Harpalani, S., Prusty, B. K., & Dutta, P. (2006). Methane/CO2 sorption modeling for coalbed methane production and CO2 sequestration. *Energy & Fuels*, **20**(4), 1591–1599.

Hawez, H. K., Sanaee, R., & Faisal, N. H. (2021). A critical review on coupled geomechanics and fluid flow in naturally fractured reservoirs. *Journal of natural gas science and engineering*, **95**, 104150.

Hedayatzadeh, M., Sarhosis, V., Gorka, T., Hámor-Vidó, M., & Kalmár, I. (2022, May). *Ground subsidence and fault reactivation during in-situ coal conversion assessed by numerical simulations.* In EGU General Assembly Conference Abstracts (pp. EGU22-11736).

Herzog, H. J. (1998, August). *Ocean sequestration of CO2: an overview.* In Proceedings of the AWMA's Second International Specialty Conference, Oct 13-15, 1998, Washington, DC.

Higgins, S. M., Goodwin, S. A., Bratton, T. R., & Tracy, G. W. (2008, September). *Anisotropic stress models improve completion design in the Baxter Shale.* In SPE Annual Technical Conference and Exhibition. OnePetro.

Hosseinzadeh, S., Haghighi, M., Salmachi, A., & Shokrollahi, A. (2024a). Carbon dioxide storage within coal reservoirs: A comprehensive review. *Geoenergy Science and Engineering*, **241**, 213198.

Hosseinzadeh, S., Abdollahi, R., Salimzadeh, S., & Haghighi, M. (2024b). Three-dimensional coupled temporal geomechanical model for fault-reactivation and surface-deformation evaluation during reservoir depletion and CO2 sequestration, securing long-term reservoir sustainability. *Sustainability*, **16**(19), 8482.

Ganguli, S. S., & Sen, S. (2020). Investigation of present-day in-situ stresses and pore pressure in the south Cambay Basin, western India: Implications for drilling, reservoir development and fault reactivation. *Marine and Petroleum Geology*, **118**, 104422.

Gardner, G. H. F., Gardner, L. W., & Gregory, A. R. (1974). Formation velocity and density—The diagnostic basics for stratigraphic traps. *Geophysics*, **39**(6), 770–780.

Gor, G. Y., Elliot, T. R., & Prévost, J. H. (2013). Effects of thermal stresses on caprock integrity during CO2 storage. *International Journal of Greenhouse Gas Control*, **12**, 300–309.

Jamaloei, B. Y. (2021). Impact of formation dilation-recompaction on cyclic steam stimulation in reservoirs with bottom water: Application of an integrated coupled reservoir-geomechanical modeling workflow. *Journal of Petroleum Science and Engineering*, **199**, 108267.

Hosseinzadeh, S., Saberi, M. R., Haghighi, M., Salmachi, A., & Salimzadeh, S. (2025a). Seismic inversion approaches for reservoir characterization: A comprehensive review. *Journal of Applied Geophysics*, **243**, 105953.

Hosseinzadeh, S., Haghighi, M., Salmachi, A., Salimzadeh, S., & Karakus, M. (2025b). Utilising 4D geomechanical modelling to evaluate formation failure and fault reactivation during hydrogen storage and production. *Australian Energy Producers Journal*, **65**(2), 1–5.

Jia, B., Tsau, J. S., & Barati, R. (2019). A review of the current progress of CO2 injection EOR and carbon storage in shale oil reservoirs. *Fuel*, **236**, 404–427.

Jung, H., Espinoza, D. N., & Hosseini, S. A. (2020). Wellbore injectivity response to step-rate CO2 injection: Coupled thermo-poro-elastic analysis in a vertically heterogeneous formation. *International Journal of Greenhouse Gas Control*, **102**, 103156.

Karimnezhad, M., Jalalifar, H., & Kamari, M. (2014). Investigation of caprock integrity for CO2 sequestration in an oil reservoir using a numerical method. *Journal of Natural Gas Science and Engineering*, **21**, 1127–1137.

Khan, S., Khulief, Y. A., & Al-Shuhail, A. A. (2018). Alleviation of pore pressure buildup and ground uplift during carbon dioxide injection into Ghawar Arab-D carbonate naturally fractured reservoir. *Environmental Earth Sciences*, **77**(12), 1–27.

Khan, S., Khulief, Y., Juanes, R., Bashmal, S., Usman, M., & Al-Shuhail, A. (2024). Geomechanical modeling of CO2 sequestration: A review focused on CO2 injection and monitoring. *Journal of Environmental Chemical Engineering*, **12**(3), 112847.

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Khandelwal, M., & Singh, T. N. (2009). Correlating static properties of coal measures rocks with P-wave velocity. *International Journal of Coal Geology*, **79**(1-2), 55–60.

Kim, S., & Hosseini, S. A. (2014). Geological CO2 storage: Incorporation of pore-pressure/stress coupling and thermal effects to determine maximum sustainable pressure limit. *Energy Procedia*, **63**, 3339–3346.

Kolawole, O., & Ispas, I. (2020). Evaluation of geomechanical properties via scratch tests: Where are we and where do we go from here?. *SN Applied Sciences*, **2**(10), 1–14.

Konstantinovskaya, E., Li, Q., Malo, M., Rivero, J. A., Faskhoodi, M. M., & Campbell, B. (2020). Strike-slip reactivation of a high-angle normal fault induced by increase of reservoir pore pressure: insight from 3D coupled reservoir-geomechanical modeling. *International Journal of Greenhouse Gas Control*, **102**, 103159.

Lambeck, K. (1987). The Perth Basin: a possible framework for its formation and evolution. *Exploration Geophysics*, **18**(1-2), 124–128.

Langhi, L., Ciftci, B., & Strand, J. (2014). Constraining fluid migration in low density data area: successful use of integrated modelling in the Southern Perth Basin. *The APPEA Journal*, **54**(2), 534–534.

Lavrov, A., Larsen, I., & Bauer, A. (2016). Numerical modelling of extended leak-off test with a pre-existing fracture. *Rock Mechanics and Rock Engineering*, **49**(4), 1359–1368.

Leclère, H., & Fabbri, O. (2013). A new three-dimensional method of fault reactivation analysis. *Journal of Structural Geology*, **48**, 153–161.

Lijith, K. P., Malagar, B. R., & Singh, D. N. (2019). A comprehensive review on the geomechanical properties of gas hydrate bearing sediments. *Marine and Petroleum Geology*, **104**, 270–285.

de Lima, A. V. S., Guimarães, L. J. N., Beserra, L. B. S., Horowitz, B., & Tueros, J. A. R. (2026). Analysis of structural integrity of geological reservoirs for CO2 storage regarding to fault reactivation: a forecasting model based on machine learning techniques. *Neural Computing and Applications*, **38**(9).

Liu, K., Hosseinzadeh, S., Safaei-Farouji, M., Liu, B., Morta, H. B., & Ostadhassan, M. (2023). Fracture toughness of organic rich shale via nanoindentation: A comparison of energy-based methods. *Geoenergy Science and Engineering*, 211695.

Liu, C. J., Wang, G. X., Sang, S. X., Gilani, W., & Rudolph, V. (2015). Fractal analysis in pore structure of coal under conditions of CO2 sequestration process. *Fuel*, **139**, 125–132.

Liu, Baoju, Jiali Qin, Jinyan Shi, Junyi Jiang, Xiang Wu, and Zhihai He. "New perspectives on utilization of CO2 sequestration technologies in cement-based materials." *Construction and Building Materials* **272** (2021): 121660.

Longuemare, P., Mainguy, M., Lemonnier, P., Onaisi, A., Gérard, C., & Koutsabeloulis, N. (2002). Geomechanics in reservoir simulation: overview of coupling methods and field case study. *Oil & gas science and technology*, **57**(5), 471–483.

Lyu, Q., Long, X., Ranjith, P. G., Tan, J., Kang, Y., & Wang, Z. (2018). Experimental investigation on the mechanical properties of a low-clay shale with different adsorption times in sub-/super-critical CO2. *Energy*, **147**, 1288–1298.

Mathur, B., Hofmann, H., Cacace, M., Hutka, G. A., & Zang, A. (2024). Thermo-hydro-mechanical simulation of cooling-induced fault reactivation in Dutch geothermal reservoirs. *Netherlands Journal of Geosciences*, **103**, e1.

Matter, J. M., Takahashi, T., & Goldberg, D. (2007). Experimental evaluation of in situ CO2-water-rock reactions during CO2 injection in basaltic rocks: Implications for geological CO2 sequestration. *Geochemistry, Geophysics, Geosystems*, **8**(2).

Mazumder, S., Karnik, A. A., & Wolf, K. H. A. (2006). Swelling of coal in response to CO2 sequestration for ECBM and its effect on fracture permeability. *Spe Journal*, **11**(03), 390–398.

McMillan, M., Will, R., Ampomah, W., Balch, R., & Czoski, P. (2019, April). *Coupled geomechanical modeling to assess cap rock integrity and mechanical fault stability: Application to Farnsworth field unit project.* In SPE Western Regional Meeting. OnePetro.

Mortazavi, A., & Maratov, T. (2024). *An investigation of fault geometric effects on fault activation mechanisms in CO2 storage.* Paper presented at the 58th US Rock Mechanics/Geomechanics Symposium, Golden, Colorado, USA. ARMA-2024-0584.

Mory, A. J., Haig, D. W., McLoughlin, S., & Hocking, R. M. (2005). Geology of the northern Perth Basin, Western Australia—a field guide. *Geology*, **9**.

Najibi, A. R., Ghafoori, M., Lashkaripour, G. R., & Asef, M. R. (2017). Reservoir geomechanical modeling: In-situ stress, pore pressure, and mud design. *Journal of Petroleum Science and Engineering*, **151**, 31–39.

Noël, C., Passelègue, F. X., Giorgetti, C., & Violay, M. (2019). Fault reactivation during fluid pressure oscillations: Transition from stable to unstable slip. *Journal of Geophysical Research: Solid Earth*, **124**(11), 10940–10953.

Nole, M., Daigle, H., Cook, A. E., & Malinverno, A. (2016). Short-range, overpressure-driven methane migration in coarse-grained gas hydrate reservoirs. *Geophysical Research Letters*, **43**(18), 9500–9508.

Noohnejad, A., Ahangari, K., & Goshtasbi, K. (2021). Comprehensive mechanical earth modeling using well data. *Innovative Infrastructure Solutions*, **6**(1), 1–13.

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Olabode, A., & Radonjic, M. (2013). Experimental investigations of caprock integrity in CO2 sequestration. *Energy Procedia*, **37**, 5014–5025.

Olierook, H. K., Barham, M., Fitzsimons, I. C., Timms, N. E., Jiang, Q., Evans, N. J., & McDonald, B. J. (2019). Tectonic controls on sediment provenance evolution in rift basins: Detrital zircon U–Pb and Hf isotope analysis from the Perth Basin, Western Australia. *Gondwana Research*, **66**, 126–142.

Olierook, H. K., Delle Piane, C., Timms, N. E., Esteban, L., Rezaee, R., Mory, A. J., & Hancock, L. (2014). Facies-based rock properties characterization for CO2 sequestration: GSWA Harvey 1 well, Western Australia. *Marine and Petroleum Geology*, **50**, 83–102.

Olierook, H. K. H., Timms, N. E., Wellmann, J. F., Corbel, S., & Wilkes, P. G. (2015). 3D structural and stratigraphic model of the Perth Basin, Western Australia: Implications for sub-basin evolution. *Australian Journal of Earth Sciences*, **62**(4), 447–467.

Ostadhassan, M., Zeng, Z., & Zamiran, S. (2012, June). *Geomechanical modeling of an anisotropic Formation-Bakken case study.* In 46th US Rock Mechanics/Geomechanics Symposium. OnePetro.

Paluszny, A., Graham, C. C., Daniels, K. A., Tsaparli, V., Xenias, D., Salimzadeh, S., … & Zimmerman, R. W. (2020). Caprock integrity and public perception studies of carbon storage in depleted hydrocarbon reservoirs. *International Journal of Greenhouse Gas Control*, **98**, 103057.

Pervukhina, M., Piane, C. D., Dewhurst, D. N., Clennell, M. B., & Bolås, H. M. N. (2013). An estimation of pore pressure in shales from sonic velocities. In *SEG Technical Program Expanded Abstracts* 2013 (pp. 2653–2658). Society of Exploration Geophysicists.

Playford, G. (2016). Mississippian palynoflora from the northern Perth Basin, Western Australia: systematics and stratigraphical and palaeogeographical significance. *Journal of Systematic Palaeontology*, **14**(9), 731–770.

Preston, B. L., & Jones, R. N. (2006). *Climate change impacts on Australia and the benefits of early action to reduce global greenhouse gas emissions (p. 41).* Canberra: CSIRO.

Qin, C., Jiang, Y., Zhou, J., Zuo, S., Chen, S., Liu, Z., … & Li, Y. (2021). Influence of supercritical CO2 exposure on water wettability of shale: Implications for CO2 sequestration and shale gas recovery. *Energy*, 122551.

Rajabi, M., Tingay, M., & Heidbach, O. (2016). The present-day state of tectonic stress in the Darling Basin, Australia: Implications for exploration and production. *Marine and Petroleum Geology*, **77**, 776–790.

Raziperchikolaee, S., Singh, V., & Kelley, M. (2020). The effect of Biot coefficient and elastic moduli stress–pore pressure dependency on poroelastic response to fluid injection: laboratory experiments and geomechanical modeling. *Greenhouse Gases: Science and Technology*, **10**(5), 980–998.

Sampath, K. H. S. M., Ranjith, P. G., & Perera, M. S. A. (2020). A comprehensive review of structural alterations in CO2-interacted coal: Insights into CO2 sequestration in coal. *Energy & Fuels*, **34**(11), 13369–13383.

Selvadurai, A. P., & Suvorov, A. P. (2016). *Thermo-poroelasticity and geomechanics.* Cambridge University Press.

Settari, A., & Walters, D. A. (2001). Advances in coupled geomechanical and reservoir modeling with applications to reservoir compaction. *Spe Journal*, **6**(03), 334–342.

Song, T., & Cawood, P. A. (2000). Structural styles in the Perth Basin associated with the Mesozoic break-up of Greater India and Australia. *Tectonophysics*, **317**(1-2), 55–72.

Steefel, C. I., Molins, S., & Trebotich, D. (2013). Pore scale processes associated with subsurface CO2 injection and sequestration. *Reviews in Mineralogy and Geochemistry Pore scale processes associated with subsurface CO2 injection and sequestration.*, **77**(1), 259–303.

Sun, Z., Garza, A., Salazar-Tio, R., Fager, A., & Crouse, B. (2021, June). *A Novel 3D Mechanical Earth Modeling of the Volve Field and Its Application to Fault Stability Analysis.* In 55th US Rock Mechanics/Geomechanics Symposium. OnePetro.

Thiercelin, M. J., & Plumb, R. A. (1994). A core-based prediction of lithologic stress contrasts in east Texas formations. *SPE Formation Evaluation*, **9**(04), 251–258.

Thompson, N., Andrews, J. S., Wu, L., & Meneguolo, R. (2022). Characterization of the in-situ stress on the Horda platform–A study from the Northern Lights Eos well. *International Journal of Greenhouse Gas Control*, **114**, 103580.

Tingay, M., Reinecker, J., & Müller, B. (2008). Borehole breakout and drilling-induced fracture analysis from image logs. *World Stress Map Project*, 1–8.

Urosevic, M., Ziramov, S., Pevzner, R., Tertyshnikov, K., Yavuz, S., Egorov, A., … & Van Gent, D. (2018, October). *CO2 storage site characterisation at Harvey, Western Australia using non-conventional seismic survey geometries.* In 14th Greenhouse Gas Control Technologies Conference Melbourne (pp. 21–26).

Van den Bogert, P. A. J., & van Eijs, R. M. H. E. (2020). Why Mohr-circle analyses may underestimate the risk of fault reactivation in depleting reservoirs. *International Journal of Rock Mechanics and Mining Sciences*, **136**, 104502.

Vidal-Gilbert, S., Nauroy, J. F., & Brosse, E. (2009). 3D geomechanical modelling for CO2 geologic storage in the Dogger carbonates of the Paris Basin. *International Journal of Greenhouse Gas Control*, **3**(3), 288–299.

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Whittaker, J. M., Williams, S. E., & Müller, R. D. (2013). Revised tectonic evolution of the Eastern Indian Ocean. *Geochemistry, Geophysics, Geosystems*, **14**(6), 1891–1909.

Xiong, Y., Fakcharoenphol, P., Winterfeld, P., Zhang, R., & Wu, Y. S. (2013, September). *Coupled geomechanical and reactive geochemical model for fluid and heat flow: application for enhanced geothermal reservoir.* In SPE reservoir characterization and simulation conference and exhibition. OnePetro.

Yang, F., Bai, B., Tang, D., Shari, D. N., & David, W. (2010). Characteristics of CO 2 sequestration in saline aquifers. *Petroleum Science*, **7**(1), 83–92.

Yin, S., & Ding, W. (2019). Evaluation indexes of coalbed methane accumulation in the strong deformed strike-slip fault zone considering tectonics and fractures: A 3D geomechanical simulation study. *Geological Magazine*, **156**(6), 1052–1068.

Zappone, A., Rinaldi, A. P., Grab, M., Wenning, Q. C., Roques, C., Madonna, C., … & Wiemer, S. (2021). Fault sealing and caprock integrity for CO 2 storage: an in situ injection experiment. *Solid Earth*, **12**(2), 319–343.

Zhang, F., & Mack, M. (2017). Integrating fully coupled geomechanical modeling with microsesmicity for the analysis of refracturing treatment. *Journal of Natural Gas Science and Engineering*, **46**, 16–25.

Zhang, Y., Langhi, L., Schaubs, P. M., Delle Piane, C., Dewhurst, D. N., Stalker, L., & Michael, K. (2015). Geomechanical stability of CO2 containment at the South West Hub Western Australia: A coupled geomechanical–fluid flow modelling approach. *International Journal of Greenhouse Gas Control*, **37**, 12–23.

Zoback, M. D., Barton, C., Brudy, M., Chang, C., Moos, D., Peska, P., … & Myrvang, A. M. (1995, February). A review of some new methods for determining the in situ stress state from observations of borehole failure with applications to borehole stability and enhanced production in the North Sea. In *Rock Stresses in the North Sea: Proceeding of the Workshop* (pp. 13–14). University of Trondheim.

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